Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 24 GB of memory. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the f1 score by dataset.

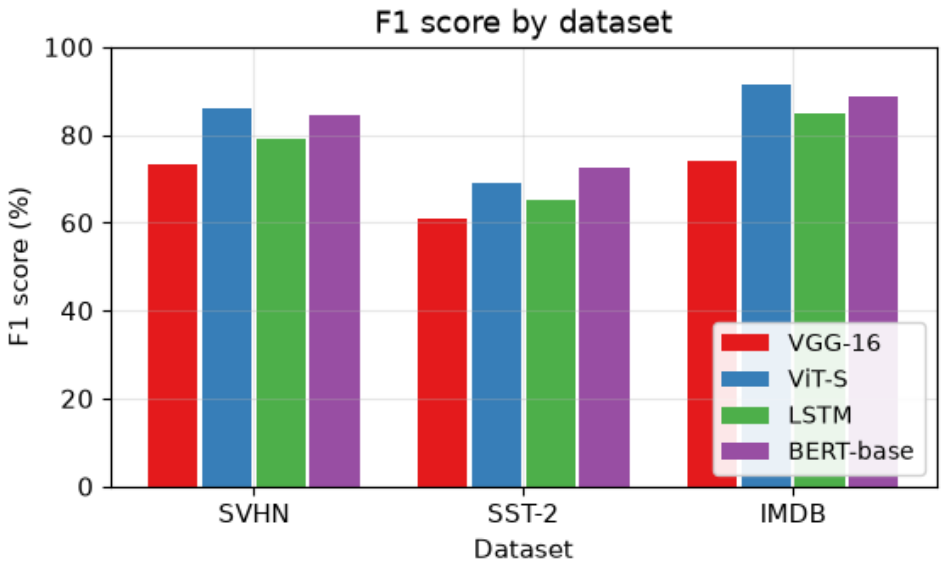


Figure 1: F1 score by dataset for the models in this experiment.

On SVHN, VGG-16 obtains a F1 score of 83.2%. As shown in Figure 1, BERT-base outperforms ViT-S by about 3 percentage points on SST-2.

Figure 2 shows the accuracy during training.

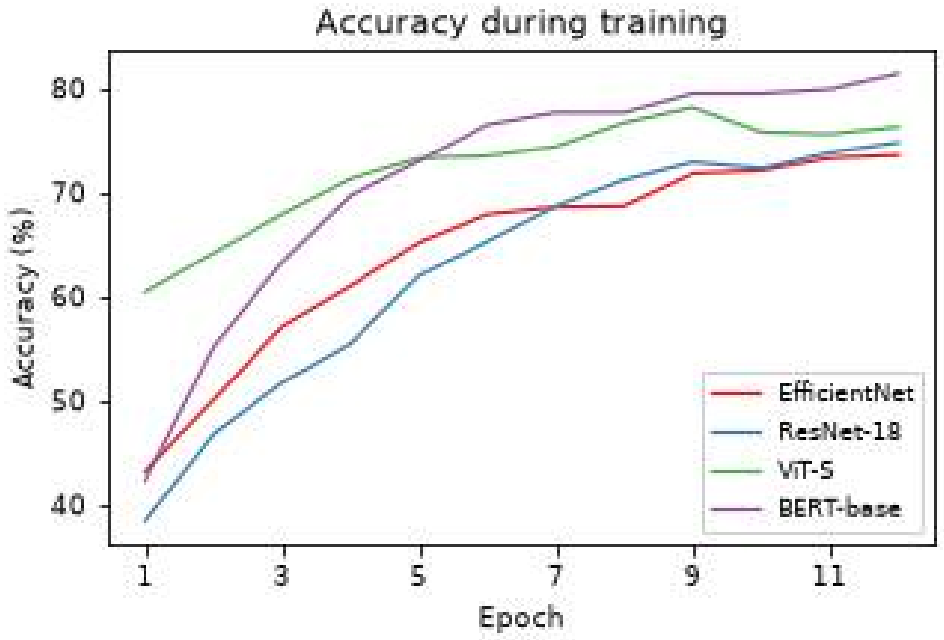


Figure 2: Accuracy during training for the models in this experiment.

BERT-base outperforms ResNet-18 by 17.9 percentage points at epoch 2 (Figure 2). As shown in Figure 2, from epoch 8 to epoch 9, the accuracy of EfficientNet improves by a relative 5%.

Figure 3 shows the validation accuracy of ViT-S by dataset.

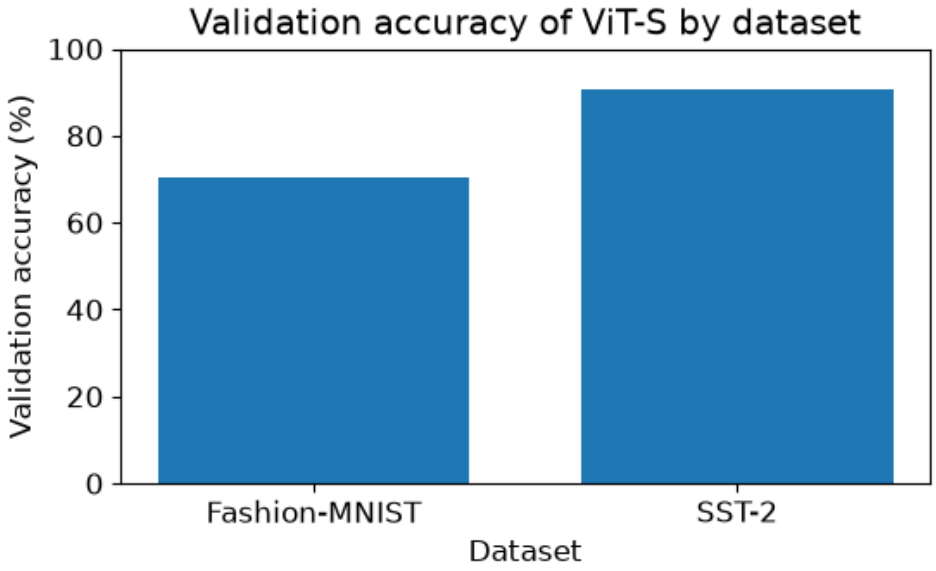


Figure 3: Validation accuracy of ViT-S by dataset for the models in this experiment.

As shown in Figure 3, ViT-S achieves 80.3% validation accuracy on Fashion-MNIST. On Fashion-MNIST, ViT-S obtains a validation accuracy of 68.3%.

3 Discussion

We note that the training was stable for all models.